



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2020-2021
MONDAY TEST

CLASS: IX
SUBJECT: BIOLOGY

FULL MARKS: 40
DATE: 5.11.20

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

1. Name the following: [1×5=5]

- i. A monocotyledonous albuminous seed.
- ii. The energy currency of the cell.
- iii. The outermost layer of the endosperm which is rich in protein.
- iv. The process by which male and the female gametes fuse.
- v. The process by which the pollen grains of one flower fall on the stigma of another flower but of the same plant.

2. Fill in the blanks: [1×5=5]

In double fertilization, one male gamete nucleus fuses with the egg cell to form a ____i____. The second male nucleus fuses with two polar nuclei to form the ____ii____. The function of ____iii____ is to direct the growth of pollen tube towards the ovule. The embryo sac consists of 3 cells at the opposite end of the micropylar end, and are known as ____iv____. After fertilization the ovules become the ____v____.

3. Differentiate between the following according to the criterion given in the bracket: [1×5=5]

- i. Herkogamy and Heterostyly (Definition)
- ii. Radicle and Plumule (Definition)
- iii. Respiration and Burning (Involvement of enzymes)
- iv. Aerobic Respiration and Anaerobic Respiration in plants (Products formed)
- v. Respiration and Photosynthesis (Place of occurrence)

4. i. Draw a neat labelled diagram of L.S. of a dicot seed. [3]

- ii. Mention one function of [2]
 - a. Seed Coat
 - b. Cotyledon

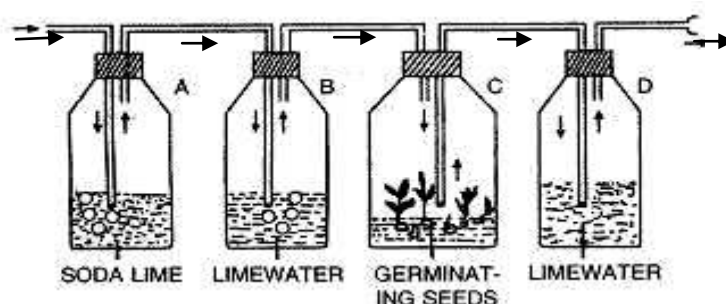
5. Give reasons for the following:

[1×5=5]

- i. Seeds sown very deep in the soil fail to germinate.
- ii. Respiration is known as a catabolic process.
- iii. Insect pollinated flowers have bright coloured petals.
- iv. Wind pollinated flowers have hanging and feathery stigma.
- v. Maize is known as a grain.

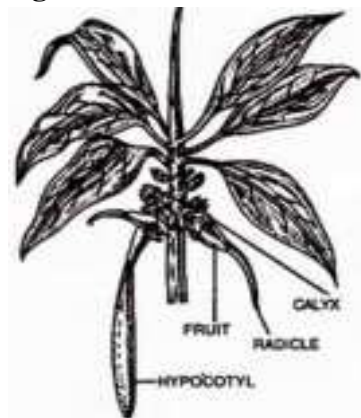
6. Answer the following questions:

[1+1+1+1+1]



- i. Why is soda lime placed in Flask 'A'?
- ii. Name the gas released from Jar "C".
- iii. Why is lime water used in the Flask D?
- iv. What do you understand by a control set up?
- v. Why do you think this experiment should be conducted in a place where there is no light?

7. Observe the diagram given below and answer the following questions: [2+1+1+1]



- i. Identify and define the type of germination shown in the above diagram.
- ii. Mention the name of two plants in which we can observe such type of germination.
- iii. Name the protective sheath of plumule and radicle.
- iv. Why is very low temperature unsuitable for germination?

8. Answer the following questions:

[1+1+2+1]



- Mention the aim of the experiment.
- Why are the seed coats peeled off before introducing them into the test tube?
- In a control set up what type of seeds do we use? Explain with reason why do we use antiseptic on those seeds which are used in the control set up?
- Explain the physiological process shown in the experiment.